

KNOWLEDGE AND REASONING



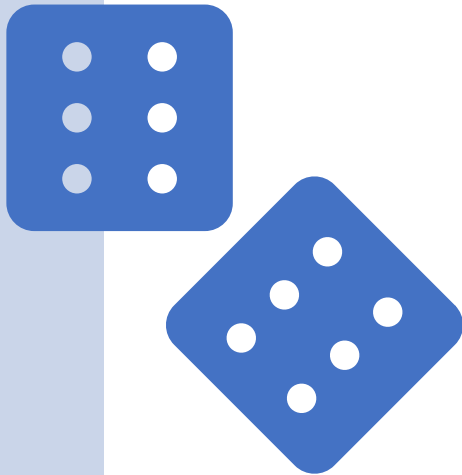
Bayesian Network

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Outline

- A brief review of probability
- Inference in uncertain domain
- Bayesian network construction

Probability: A brief review



Basic concepts in probability

- Let $X = (X_1, \dots, X_n)$ be random variables.
- Suppose X can be partitioned into the sets, A and B .
 - E.g., $A = (X_1; X_3)$ and $B = (X_2; X_4; X_5)$ if $n = 5$
- Joint distribution: $\mathbf{P}(X) = P(X_1, \dots, X_n)$
- Marginal distribution: $P(A) = \sum_b P(A, B = b)$
- Conditional distribution: $P(A \mid B = b) \propto P(A, B = b)$

Basic concepts: An example



- Random variables: sunshine $S \in \{0,1\}$ and rain $R \in \{0,1\}$

- Joint distribution:

s	r	$P(S = s, R = r)$
0	0	0.20
0	1	0.08
1	0	0.70
1	1	0.02

- Marginal distribution:

s	$P(S = s)$
0	0.28
1	0.72

(aggregate rows)

- Conditional distribution:

s	$P(S = s R = 1)$
0	0.8
1	0.2

(select rows, normalize)

Probabilistic inference task

- **Probabilistic inference** involves determining the distribution of unobserved variables based on known information..

- **Random variables**: unknown quantities in the world

$$X = (S, R, T, A)$$

- **Context**: Suppose that observed events are $T = 1$ and $A = 1$, and you are interested in querying R .

- **Represented in symbols**:

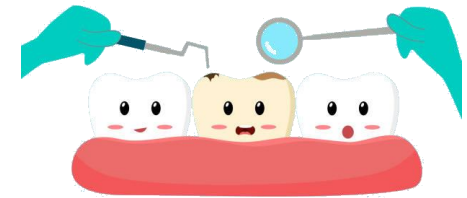
$$\mathbf{P}(\underbrace{R}_{\text{query}} \mid \underbrace{T = 1, A = 1}_{\text{condition}})$$

(S is marginalized out)

Full joint probability distribution

- Full joint probability distribution (FJPD) can represent the probabilities of all possible worlds.

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576



$$P(\textit{Toothache}) = \langle 0.2, 0.8 \rangle$$

$$P(\neg \textit{cavity} \mid \textit{toothache})$$

$$= \frac{P(\neg \textit{cavity} \wedge \textit{toothache})}{P(\textit{toothache})} = \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064}$$

$$P(\textit{cavity} \vee \textit{toothache}) = 0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064$$

Full joint probability distribution

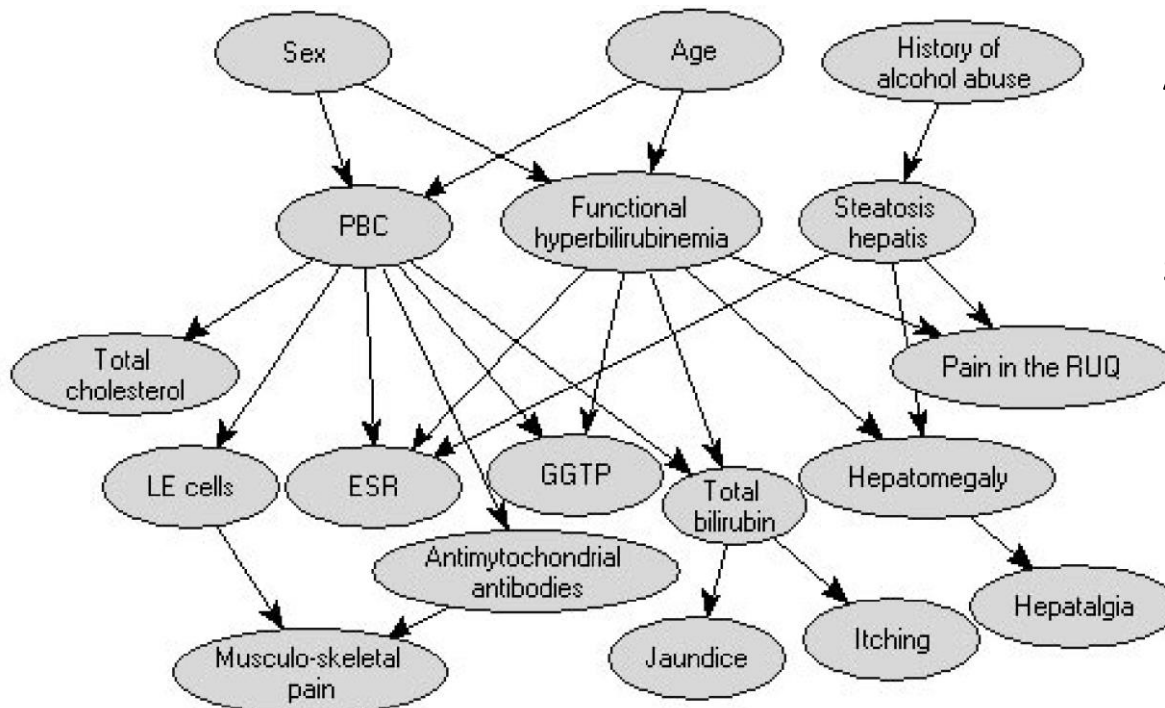
- We can answer every question about a domain by the joint distribution because every event is a sum of sample points.
- However, a plain FJPD may become intractably large as the number of variables grows.
 - Defining probabilities for possible worlds is unnatural and tedious.
- (Conditional) independence relationships among variables can greatly reduce the number of probabilities required.

Inference in uncertain domain



Bayesian networks

- **Bayesian networks** (BN) can represent essentially, and in many cases very concisely, any FJPD.
- Other terms: Belief network, probabilistic network, causal network, knowledge map

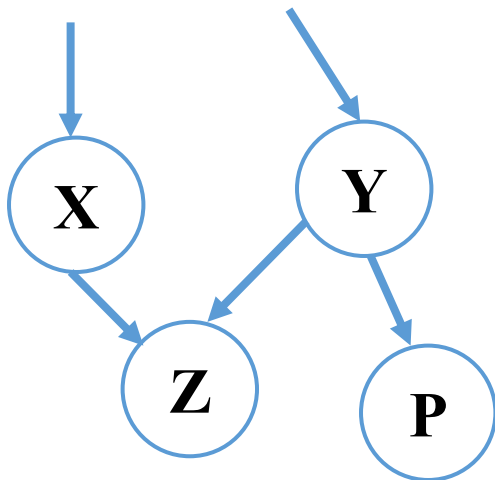


A simplified fragment of the Hepar-BN Bayesian network model, which diagnose a small set of hepatic disorders.

[Source](#)

Network topology: Nodes and links

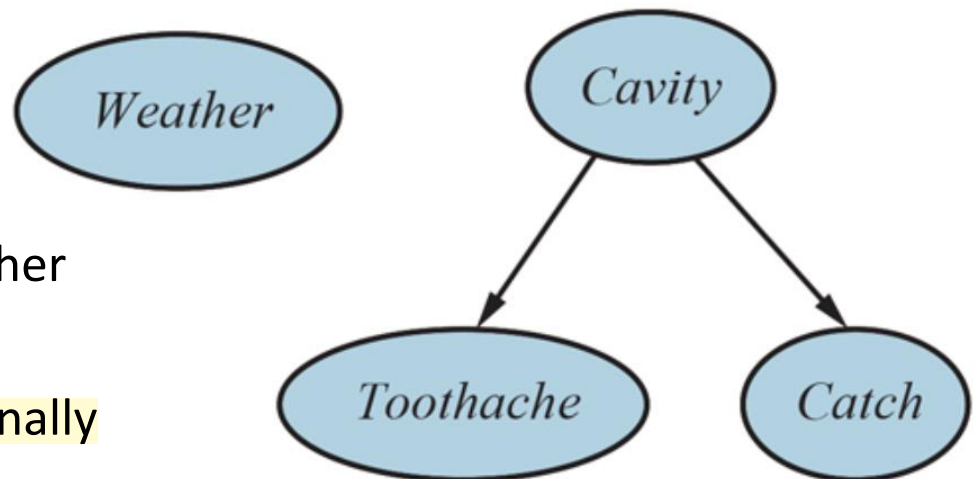
- A Bayesian network is a **directed acyclic graph (DAG)**.
- Each **node X_i** in the network presents a **random variable**.
 - It associates with **$P(X_i | X_{Parent(i)})$** , which quantifies the **effect of the parents $X_{Parent(i)}$** on the node.
- There are **directed links** connects **pairs of nodes**.
 - If there is a link from node X to node Y , X is a parent of Y .



- X and Y are the parents of Z
- Y is the parent of P
- No direct dependency between Z and P
- **No loops/cycles**

Network topology: Relationships

- The **network topology** defines the **conditional independence relationships** that hold in the domain.
 - $X \rightarrow Y$: X has a **direct influence** on Y , suggesting that causes should be parents of effects.
- A **domain expert** will decide what direct influences exist in that domain.



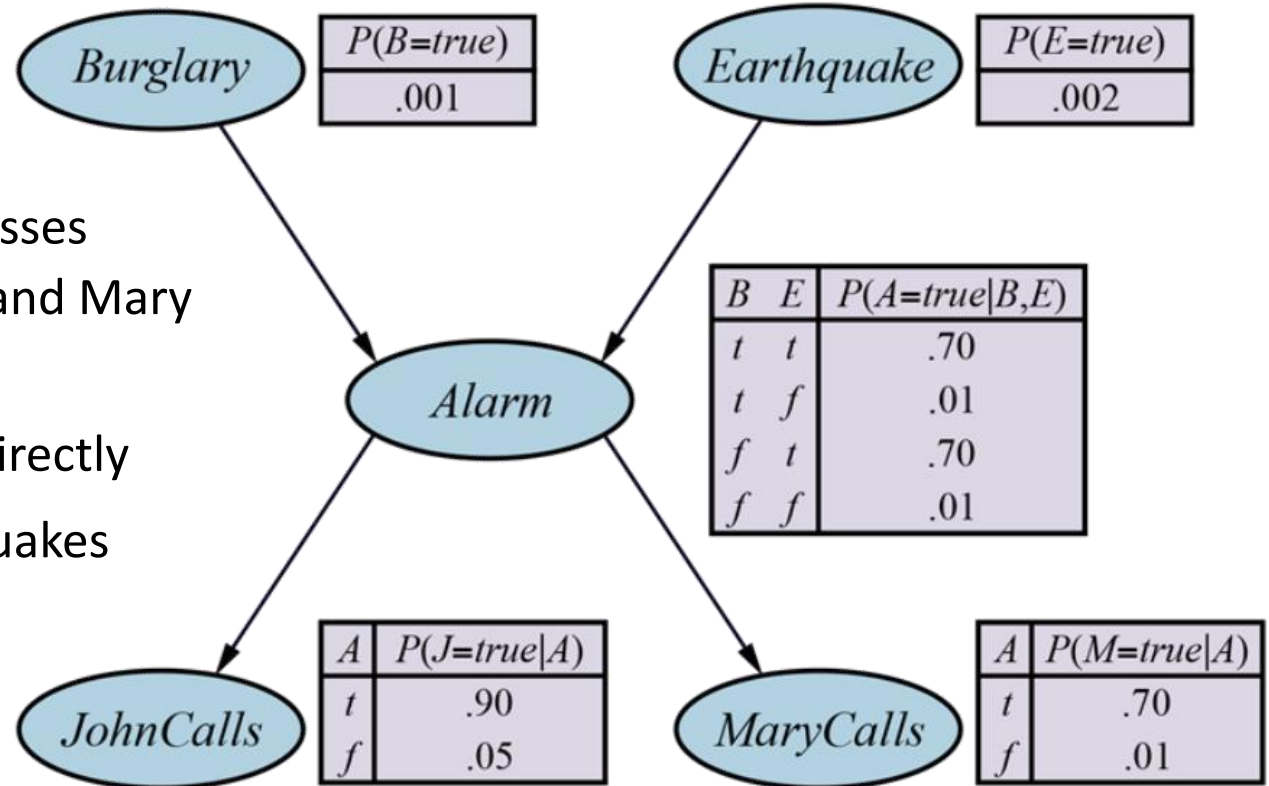
Weather is independent of the other three variables.

Toothache and Catch are **conditionally independent**, given Cavity

Network topology: An example

The network topology shows that

- Burglary and earthquakes directly affect the probability of the alarm's going off
- Whether John and Mary call depends only on the alarm.



The network thus expresses assumptions that John and Mary do not

- Perceive burglaries directly
- Notice minor earthquakes
- Confer before calling

Network topology: Conditional probability

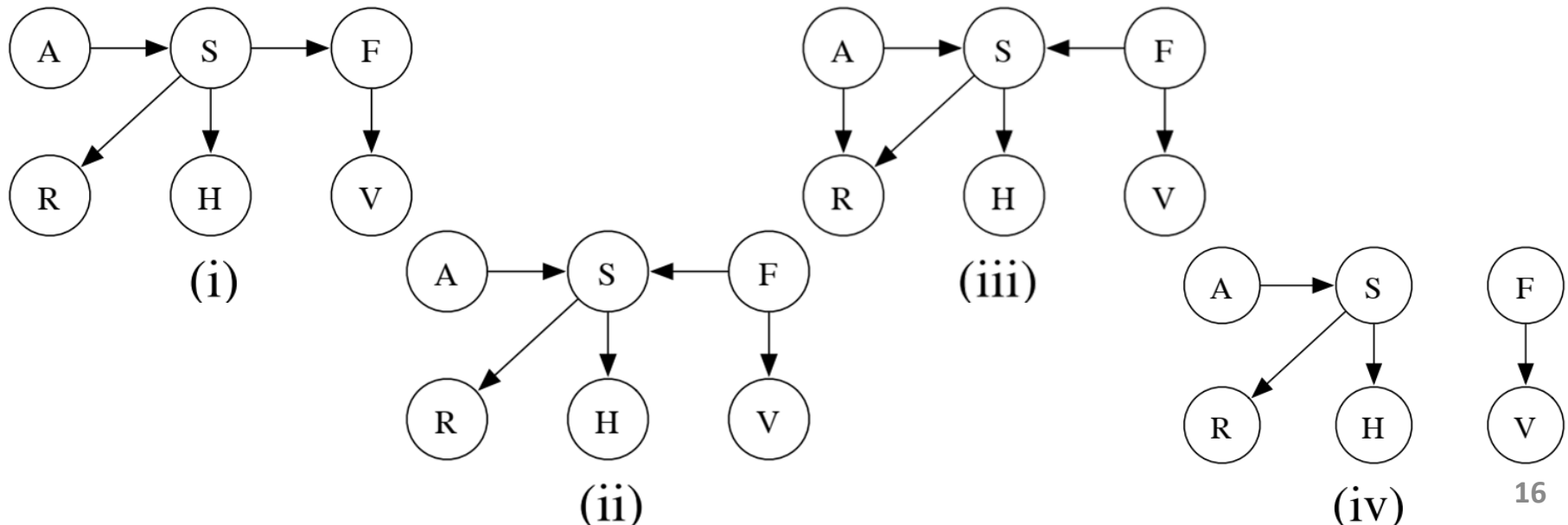
- Consider a Bayesian network of discrete variables.
- **Conditional probability table** (CPT) shows the conditional probability distribution for each variable, given its parents.
- Each **row** contains the conditional probability of each node value for a **conditioning case** → the entries must sum to 1.
 - **Conditioning case**: a possible combination of parent nodes' values.
- A CPT for a Boolean variable X with k Boolean parents has 2^k independently specifiable probabilities.
 - X needs only the probability of true value p , since the other is $1 - p$.

Network topology: Conditional probability

- The probabilities summarize a **potentially infinite set of circumstances** in which an event does (not) happen.
- For example, $P(\neg Alarm \mid Burlary, Earthquake) = 0.3$
 - The alarm might fail to go off (high humidity, power failure, dead battery, a dead mouse stuck inside, etc.)
- $P(\neg John \mid Alarm) = 0.1$ and $P(\neg Mary \mid Alarm) = 0.3$
 - John or Mary might fail to call and report it (on vacation, negligent, passing helicopter, etc.).

Quiz 01: Bayesian net – Snuffles

- Assume there are two types of conditions: (S)inus congestion and (F)lu. Sinus congestion is caused by (A)llergy or the flu. There are three observed symptoms for these conditions: (H)eadache, (R)unny nose, and fe(V)er. Runny nose and headaches are directly caused by sinus congestion (only), while fever comes from having the flu (only). For example, allergies cause runny noses indirectly.
- Assume each variable is Boolean. Consider the four Bayesian networks shown. Choose the one which models the domain best. Explain why the others do not.



Quiz 02: Bayesian net – Rainy city

- Kim lives in London, and every morning, before she goes to school, she looks at the sky to see if it is cloudy. If it is, she usually brings an umbrella (about 80% of the time). Otherwise, there's still a 10% chance that she gets the umbrella. If it is cloudy in the morning, there's a 70% chance it will rain in the afternoon. However, if the morning sky is clear, there's still a 30% possibility of rain in the afternoon. In London, 60% of the mornings are cloudy. On the way home in the afternoon, Kim may get wet. If it is raining and she brought an umbrella, there's only a 20% chance of getting wet. However, if she forgets the umbrella, it is sure that she will get wet. If it's not raining, there's still a 10% chance that Kim gets wet, and this is regardless of whether she has an umbrella or not.
- Construct the Bayesian network for the above problem (assuming only dependencies described in the text) using the following Boolean variables.
 - C: "It is cloudy in the morning."
 - U: "Kim brings an umbrella."
 - R: "There is rain in the afternoon."
 - W: "Kim gets wet on the way home."

Bayesian networks: A definition

- A **joint distribution over X** is defined as a **product of local conditional distributions**, one for each node.

$$\mathbf{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \mathbf{P}(x_i \mid x_{Parent(i)})$$

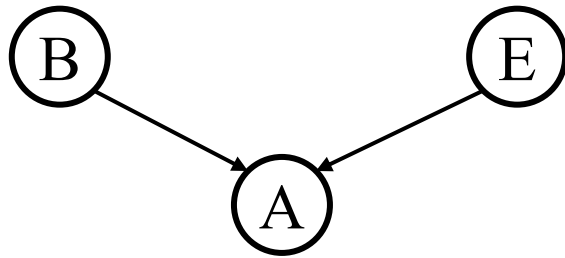
- $x_{Parent(i)}$ is an assignment to the parents of X_i (possibly none).
- **Key idea:** **locally normalized**

All factors (local conditional distributions) satisfy:

$$\sum_{x_i} p(x_i \mid x_{Parent(i)}) = 1 \text{ for each } x_{Parent(i)}$$

Properties: Consistency of sub-BNs

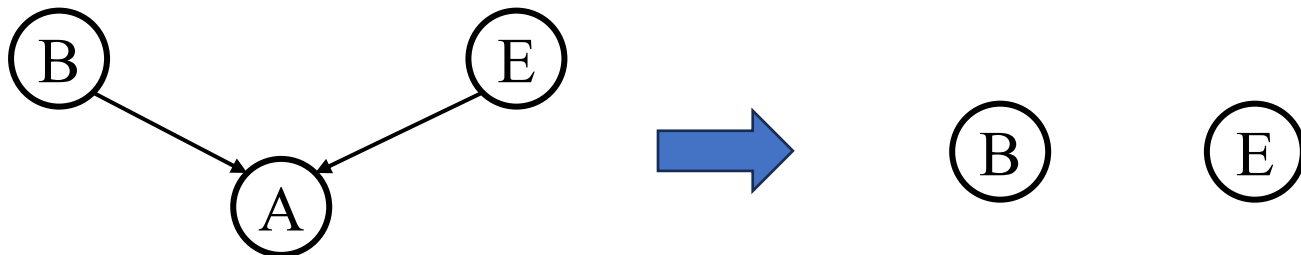
- If we marginalize a variable, a sub-BN is defined with one fewer variable, and the same local conditional probabilities.



$$\begin{aligned} P(B = b, E = e) &= \sum_a P(B = b, E = e, A = a) \\ &= P(b)P(e) \sum_a P(a | b, e) \\ &= P(b)P(e) \end{aligned}$$

- **Key idea:** Marginalization

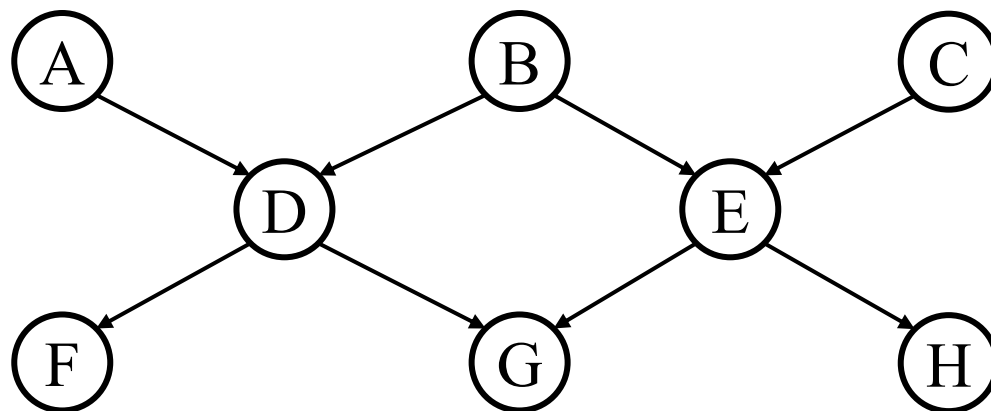
Marginalizing a leaf node yields a BN without that node.



Properties: Consistency of local conditionals

- **Key idea:** Local condition distributions

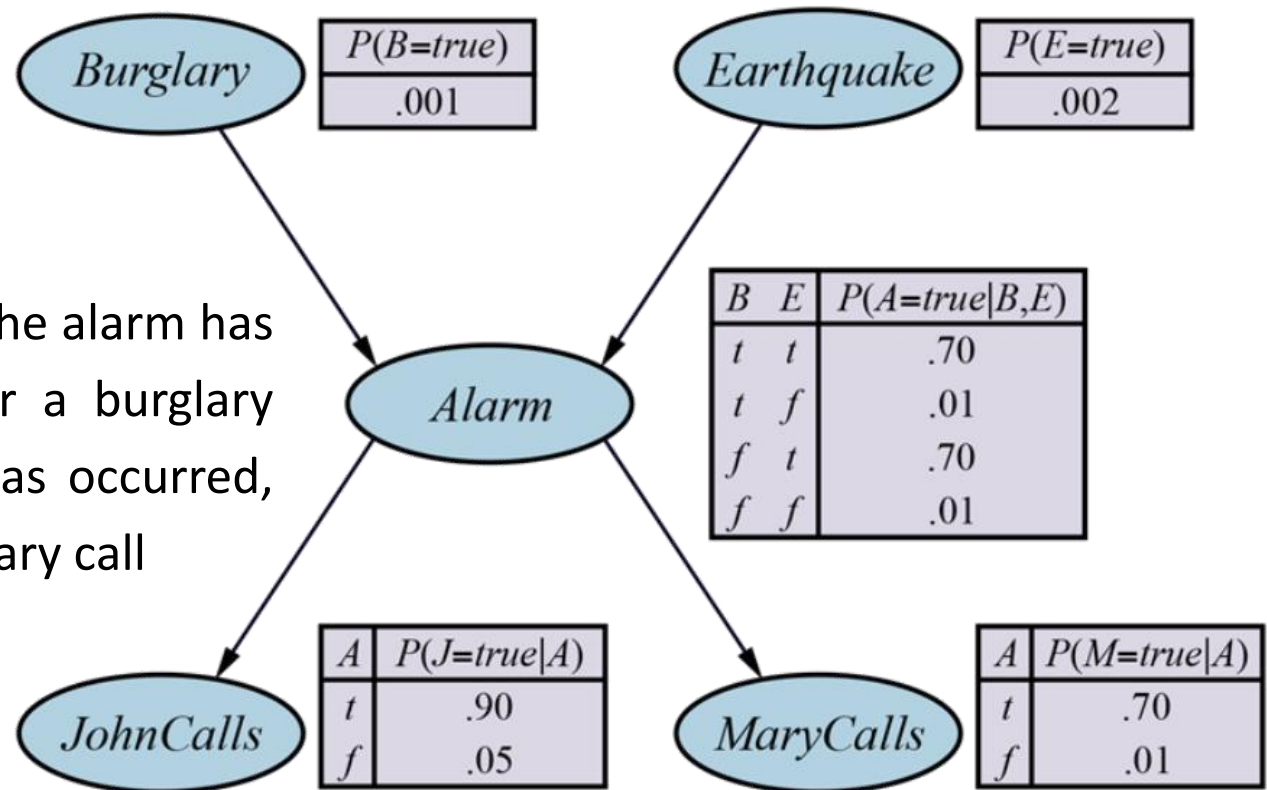
Local conditional distributions (factors) are the true conditional distributions.



$$P(D = d \mid A = a, B = b) = P(d \mid a, b)$$

BN inference: An example

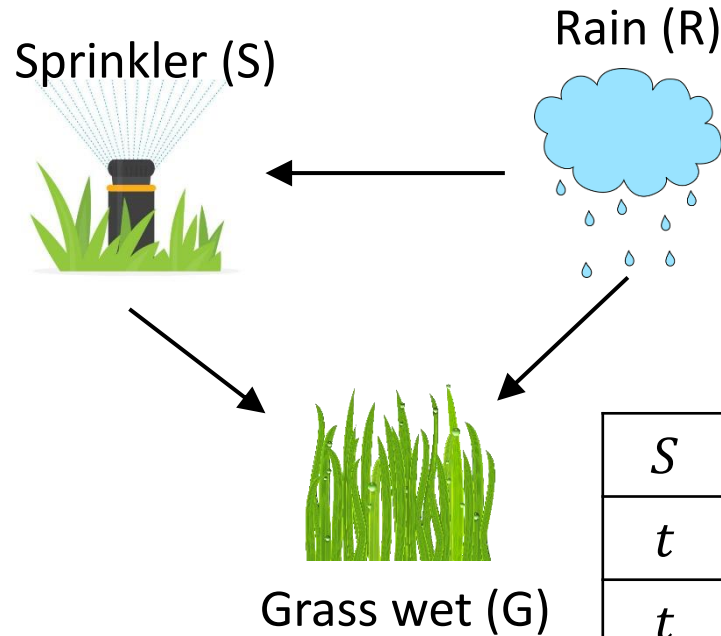
- The probability that the alarm has sounded, but neither a burglary nor an earthquake has occurred, and both John and Mary call



$$\begin{aligned}
 P(j, m, a, \neg b, \neg e) &= P(j | a) P(m | a) P(a | \neg b, \neg e) P(\neg b) P(\neg e) \\
 &= 0.90 \times 0.70 \times 0.01 \times 0.999 \times 0.998 = 0.00628
 \end{aligned}$$

BN inference: The wet grass example

R	$P(S)$
t	0.01
f	0.4



$P(R)$
0.2

S	R	$P(\textcolor{red}{S})$
t	t	0.99
t	f	0.9
f	t	0.8
f	f	0.0

BN inference: The wet grass example

- What is the probability that it is raining, given the grass is wet?

$$P(R = t | G = t) = \frac{P(G = t, R = t)}{P(G = t)} = \frac{\sum_{S \in \{T, F\}} P(G = t, S, R = t)}{\sum_{S, R \in \{T, F\}} P(G = t, S, R)}$$

- Use the expansion for the joint probability function $P(G, S, R)$ and the conditional probabilities from the CPTs stated in the diagram

$$\begin{aligned} P(G = t, S = t, R = t) &= P(G = t | S = t, R = t) P(S = t | R = t) P(R = t) \\ &= 0.99 \times 0.01 \times 0.2 = 0.00198 \end{aligned}$$

- The numerical results (subscripted by the associated variable values) are

$$\begin{aligned} P(R = t | G = t) &= \frac{0.00198_{TTT} + 0.1584_{TFT}}{0.00198_{TTT} + 0.288_{TTF} + 0.1584_{TFT} + 0.0_{TFF}} \\ &= \frac{891}{2491} \approx 35.77\% \end{aligned}$$

Inference by enumeration: Notations

- Let X be the query variable, $\mathbf{E}$ be the set of evidences $E_1, \dots, E_m$, and $\mathbf{Y}$ be the hidden variables, $Y_1, \dots, Y_l$.
- Thus, the complete set of variables is $\mathbf{X} = \{X\} \cup \mathbf{E} \cup \mathbf{Y}$
- A typical query asks for the posterior probability $P(X | e)$ for some observed event(s) e , which can be computed from BN.

$$P(X | e) = \alpha P(X, e) = \alpha \sum_y P(X, e, y)$$

- The constant term α is usually simplified during calculation.

Inference by enumeration: An example

- Consider the following query

$$\mathbf{P}(\textit{Burglary} \mid \textit{JohnCalls} = \textit{true}, \textit{MaryCalls} = \textit{true})$$

- The hidden variables are *Earthquake* and *Alarm*.
- Using initial letters for the variables, we have

$$\mathbf{P}(B \mid j, m) = \alpha \mathbf{P}(B, j, m) = \alpha \sum_e \sum_a \mathbf{P}(B, j, m, e, a)$$

- For simplicity, we do this for *Burglary* = *true*.

$$P(b \mid j, m) = \alpha \sum_e \sum_a P(b) P(e) P(a \mid b, e) P(j \mid a) P(m \mid a)$$

- Worst-case complexity:** $O(n2^n)$ for a network of n Boolean variables.
 - That is, we must sum out almost all the variables.

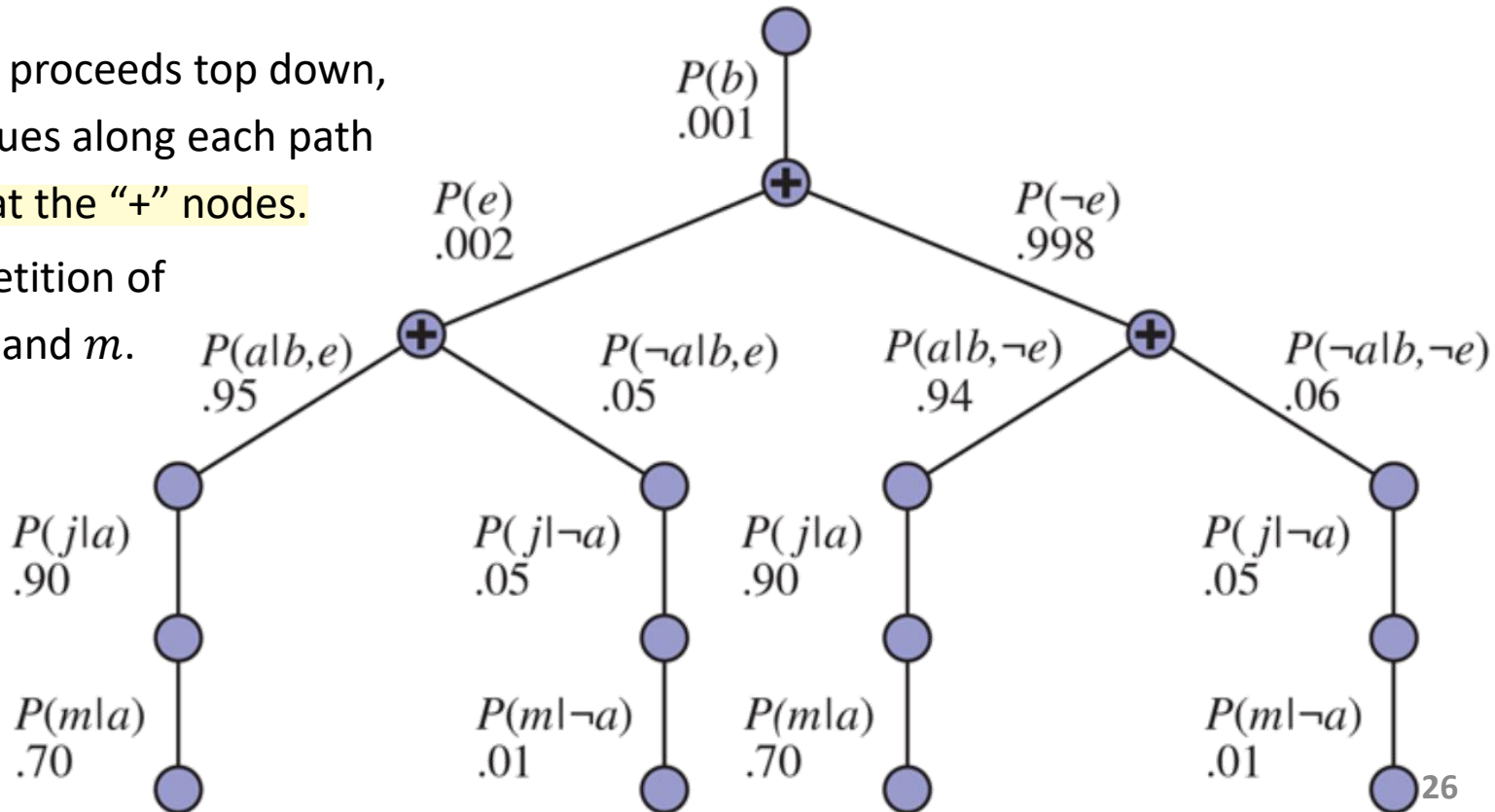
Inference by enumeration

- An improvement can be obtained as follows.

$$P(b \mid j, m) = \alpha P(b) \sum_e P(e) \sum_a P(a \mid b, e) P(j \mid a) P(m \mid a)$$

The evaluation proceeds top down, multiplying values along each path and summing at the “+” nodes.

Notice the repetition of the paths for j and m .



Inference by enumeration: Pseudocode

function ENUMERATION-ASK($X, \mathbf{e}, bn$) **returns** a distribution over X
inputs: X , the query variable, $\mathbf{e}$, observed values for variables $\mathbf{E}$
 bn , a Bayes net with variables $vars$
 $Q(X) \leftarrow$ a distribution over X , initially empty
for each value x_i of X **do**
 $Q(x_i) \leftarrow$ ENUMERATE-ALL($vars, \mathbf{e}_{x_i}$) where $\mathbf{e}_{x_i}$ is $\mathbf{e}$ extended with $X = x_i$
return NORMALIZE($Q(X)$)

function ENUMERATE-ALL($vars, \mathbf{e}$) **returns** a real number
if EMPTY?($vars$) **then return** 1.0
 $V \leftarrow$ FIRST($vars$)
if V is an evident variable with value v in $\mathbf{e}$
 then return $P(v \mid \text{parents}(V)) \times \text{ENUMERATE-ALL}(\text{REST}(vars), \mathbf{e})$
 else return $\sum_v P(v \mid \text{parents}(V)) \times \text{ENUMERATE-ALL}(\text{REST}(vars), \mathbf{e}_v)$
 where $\mathbf{e}_v$ is $\mathbf{e}$ extended with $V = v$

Inference by enumeration: Complexity

- The **space complexity** of ENUMERATION-ASK is **only linear in the number of variables**.
 - The algorithm sums over the FJPD without constructing it explicitly.
- The **time complexity** for a network with n Boolean variables (not counting the evidence variables) is **always $O(2^n)$**
 - Better than the $O(n2^n)$ for the simple approach, but still rather grim.
- There are still **repeated subexpressions** to be evaluated.
 - E.g., $P(j \mid a)P(m \mid a)$ and $P(j \mid \neg a)P(m \mid \neg a)$ are computed twice, once for each value of e .

Variable elimination algorithm

- **Dynamic programming:** do the calculation once and save the results for later use
 - Evaluate expressions in *right-to-left* order (i.e., bottom up in the tree)
 - Store intermediate results and do summations over each variable only for portions of the expression that depend on the variable.

```
function ELIMINATION-ASK( $X, \mathbf{e}, bn$ ) returns a distribution over  $X$   
  inputs:  $X$ , the query variable  $\mathbf{e}$ , observed values for variables  $\mathbf{E}$   
            $bn$ , a Bayesian network with variable  $vars$   
   $factors \leftarrow []$   
  for each  $V$  in ORDER( $vars$ ) do  
     $factors \leftarrow [MAKE-FACTOR(V, \mathbf{e}) \mid factors]$   
    if  $V$  is a hidden variable then  $factors \leftarrow SUM-OUT(V, factors)$   
  return NORMALIZE(POINTWISE-PRODUCT( $factors$ ))
```

Variable elimination: Factorization

- Consider the burglary network. We evaluate the following

$$P(B \mid j, m) = \alpha \underbrace{P(B)}_{\text{factor}} \sum_e \underbrace{P(e)}_{\mathbf{f}_1(B)} \sum_a \underbrace{P(a \mid B, e)}_{\mathbf{f}_2(E)} \underbrace{P(j \mid a)}_{\mathbf{f}_3(A, B, E)} \underbrace{P(m \mid a)}_{\mathbf{f}_4(A)} \underbrace{P(m \mid a)}_{\mathbf{f}_5(A)}$$

- $\mathbf{f}_4(A) = \begin{pmatrix} P(j \mid a) \\ P(j \mid \neg a) \end{pmatrix} = \begin{pmatrix} 0.90 \\ 0.05 \end{pmatrix}$, $\mathbf{f}_5(A) = \begin{pmatrix} P(m \mid a) \\ P(m \mid \neg a) \end{pmatrix} = \begin{pmatrix} 0.70 \\ 0.01 \end{pmatrix}$
- $\mathbf{f}_3(A, B, E)$ is a $2 \times 2 \times 2$ matrix
- In terms of factors, the query expression is written as

$$P(B \mid j, m) = \alpha \mathbf{f}_1(B) \times \sum_e \mathbf{f}_2(E) \times \sum_a \mathbf{f}_3(A, B, E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A)$$

where $\times$ is the pointwise product operation.

Variable elimination: Factorization

- First, we sum out A from the product of $\mathbf{f}_3$, $\mathbf{f}_4$, and $\mathbf{f}_5$.

$$\begin{aligned}\mathbf{f}_6(B, E) &= \sum_a \mathbf{f}_3(A, B, E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A) \\ &= \mathbf{f}_3(a, B, E) \times \mathbf{f}_4(a) \times \mathbf{f}_5(a) + \mathbf{f}_3(\neg a, B, E) \times \mathbf{f}_4(\neg a) \times \mathbf{f}_5(\neg a)\end{aligned}$$

Now we are left with $P(B \mid j, m) = \alpha \mathbf{f}_1(B) \times \sum_e \mathbf{f}_2(E) \times \mathbf{f}_6(B, E)$

- Next, we sum out E from the product of $\mathbf{f}_2$ and $\mathbf{f}_6$

$$\mathbf{f}_7(B) = \sum_e \mathbf{f}_2(E) \times \mathbf{f}_6(B, E) = \mathbf{f}_2(e) \times \mathbf{f}_6(B, e) + \mathbf{f}_2(\neg e) \times \mathbf{f}_6(B, \neg e)$$

- The following expression can be evaluated by taking pointwise product and normalizing the result.

$$P(B \mid j, m) = \alpha \mathbf{f}_1(B) \times \mathbf{f}_7(B)$$

Variable elimination: Operators

- The pointwise product of two factors $\mathbf{f}_1$ and $\mathbf{f}_2$ gives a new factor $\mathbf{f}$ whose
 - Variables are the union of the variables in $\mathbf{f}_1$ and $\mathbf{f}_2$
 - Elements are given by the product of the corresponding elements in the two factors.
- Suppose the two factors have variables $Y_1, \dots, Y_k$ in common.
- Then, $\mathbf{f}_3(X_1 \dots X_j, Y_1 \dots Y_k, Z_1 \dots Z_l) = \mathbf{f}_1(X_1 \dots X_j, Y_1 \dots Y_k) \times \mathbf{f}_2(Y_1 \dots Y_k, Z_1 \dots Z_l)$

A	B	$\mathbf{f}_1(A, B)$	B	C	$\mathbf{f}_2(B, C)$	A	B	C	$\mathbf{f}_3(A, B, C)$
T	T	.3	T	T	.2	T	T	T	$.3 \times .2 = .06$
T	F	.7	T	F	.8	T	T	F	$.3 \times .8 = .24$
F	T	.9	F	T	.6	T	F	T	$.7 \times .6 = .42$
F	F	.1	F	F	.4	T	F	F	$.7 \times .4 = .28$
if all the variables are binary, $\mathbf{f}_1$ and $\mathbf{f}_2$ have 2^{j+k} and 2^{k+l} entries, respectively, and their product has 2^{j+k+l} .						F	T	T	$.9 \times .2 = .18$
						F	T	F	$.9 \times .8 = .72$
						F	F	T	$.1 \times .6 = .06$
						F	F	F	$.1 \times .4 = .04$

Variable elimination: Operators

- Summing out a variable from a product of factors is done by adding up the submatrices formed by fixing the variable to each of its values in turn.

$$\mathbf{f}(B, C) = \sum_a \mathbf{f}_3(A, B, C) = \mathbf{f}_3(a, B, C) + \mathbf{f}_3(\neg a, B, C)$$

- Notice that any factor that does not depend on the variable to be summed out can be moved outside the summation.

$$\sum_e \mathbf{f}_2(E) \times \mathbf{f}_3(A, B, E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A) = \mathbf{f}_4(A) \times \mathbf{f}_5(A) \times \sum_e \mathbf{f}_2(E) \times \mathbf{f}_3(A, B, E)$$

Variable elimination: Ordering

- Every choice of ordering yields a valid algorithm
- Different orderings cause different intermediate factors to be generated during the calculation.

$$\begin{aligned} P(B \mid j, m) &= \alpha \mathbf{f}_1(B) \times \sum_e \mathbf{f}_2(E) \times \sum_a \mathbf{f}_3(A, B, E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A) \\ &= \alpha \mathbf{f}_1(B) \times \sum_a \mathbf{f}_4(A) \times \mathbf{f}_5(A) \times \sum_e \mathbf{f}_2(E) \times \mathbf{f}_3(A, B, E) \end{aligned}$$

$\mathbf{f}_6$

- A heuristic for variable elimination

Every variable that is not an ancestor of a query variable or evidence variable is irrelevant to the query.

Variable elimination: An example

- Consider the following network. Calculate $P(B \mid \neg c)$.

- $$P(B \mid \neg c) = \alpha \underbrace{P(\neg c \mid B)}_{\text{factor } f_1(B)} \times \sum_a \underbrace{P(B \mid a)}_{f_2(A, B)} \times \underbrace{P(a)}_{f_3(A)}$$

- Irrelevant variable: D . Observed variable: $C = \neg c$.

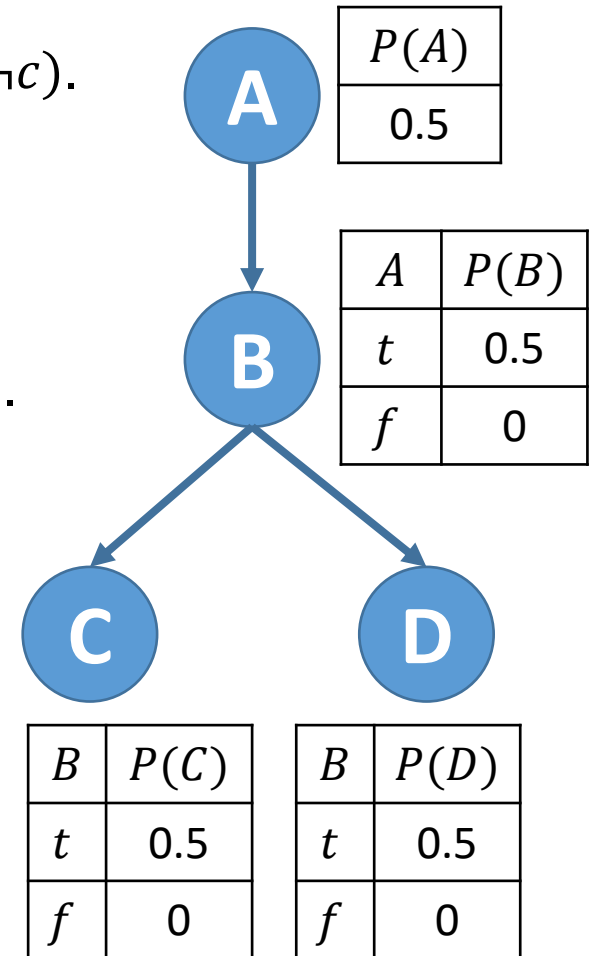
- Sum out A to have $f_4(B) = \sum_a P(B \mid a) \times P(a)$

- Join f_1 and f_4 : $f_5(B, \neg c) = f_1(B) \times f_4(B)$

- Finally, we have $P(B \mid \neg c) = \alpha f_5(B, \neg c)$

- Assume that $B = b$. Normalize for B :

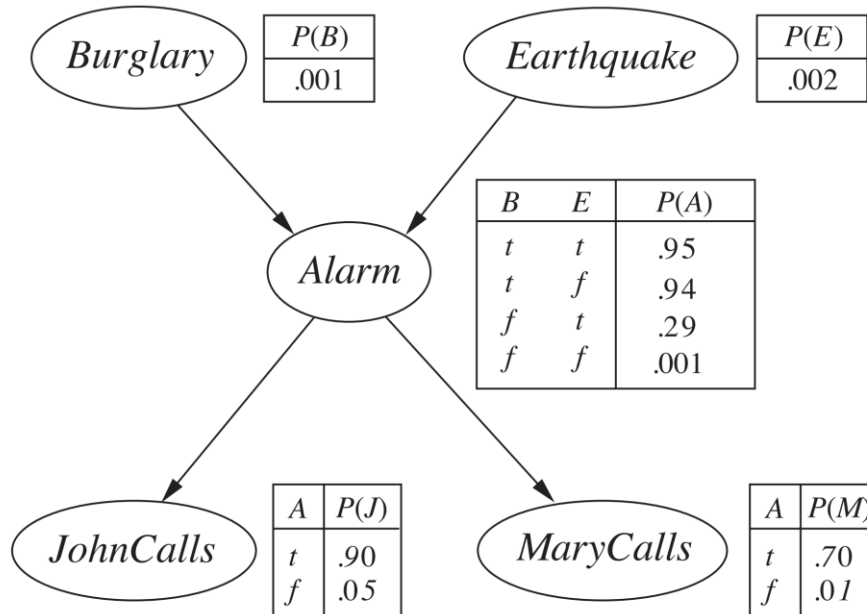
$$P(b \mid \neg c) = \frac{f_5(b, \neg c)}{f_5(b, c) + f_5(\neg b, \neg c)}$$



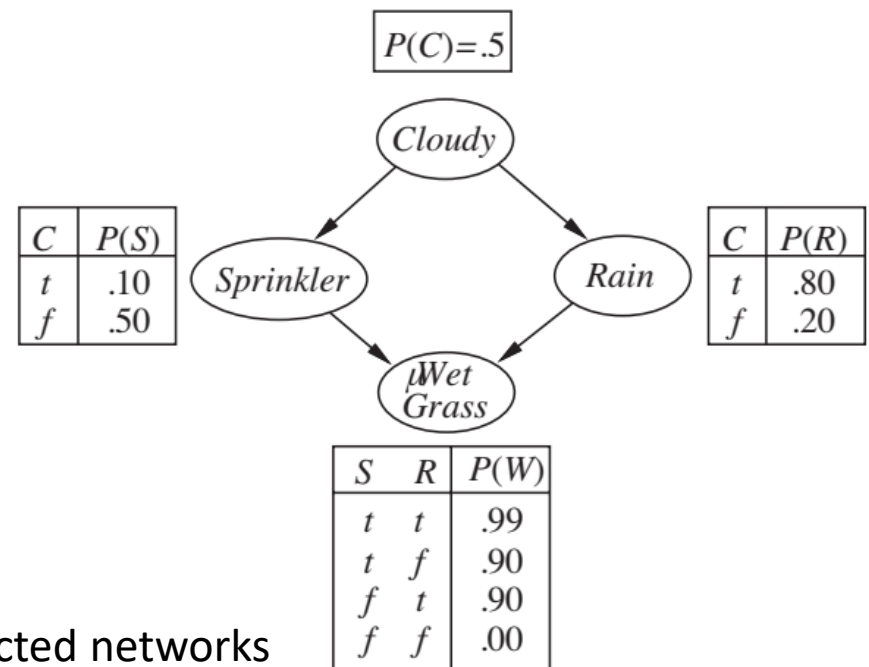
The complexity of exact inference

- The complexity of exact inference depends strongly on the structure of the network.
- Singly connected networks or polytrees: Linear time and space complexity to the network size.
 - The size is defined as the number of CPT entries.
 - If the number of parents of each node is bounded by a constant, the complexity will also be linear in the number of nodes.
- Multiply connected networks: Exponential time and space complexity in the worst case, even when the number of parents per node is bounded
 - Inference in Bayesian networks is NP-hard.

The complexity of exact inference



Singly connected networks or polytrees:
there is **at most one undirected path**
between any two nodes in the network



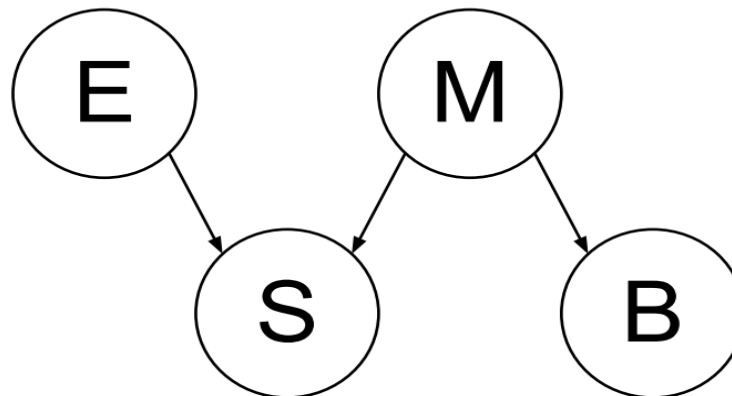
Multiply connected networks

Quiz 02: Inference in Bayesian networks

- A smell of sulfur (S) can be caused either by rotten eggs (E) or as a sign of the doom brought by the Mayan Apocalypse (M). The Mayan Apocalypse also causes the oceans to boil (B).
- The Bayesian network and corresponding CPTs are shown below.

$P(E)$	
$+e$	0.4
$-e$	0.6

$P(S E, M)$			
$+e$	$+m$	$+s$	1.0
$+e$	$+m$	$-s$	0.0
$+e$	$-m$	$+s$	0.8
$+e$	$-m$	$-s$	0.2
$-e$	$+m$	$+s$	0.3
$-e$	$+m$	$-s$	0.7
$-e$	$-m$	$+s$	0.1
$-e$	$-m$	$-s$	0.9



$P(M)$	
$+m$	0.1
$-m$	0.9

$P(B M)$		
$+m$	$+b$	1.0
$+m$	$-b$	0.0
$-m$	$+b$	0.1
$-m$	$-b$	0.9

Quiz 02: Inference in Bayesian networks

- Write down the joint probability distribution from the given network

$$P(E, S, M, B) =$$

- Compute the following entry from the joint distribution

$$P(+e, +s, -m, +b) =$$

- What is the probability that the oceans boil?

$$P(+b) =$$

- What is the probability that the Mayan Apocalypse is occurring, given that the oceans are boiling?

$$P(+m \mid +b) =$$

- What is the probability that rotten eggs are present, given that the Mayan Apocalypse is occurring?

$$P(+e \mid +m) =$$

- What is the probability that the Mayan Apocalypse is occurring, given that there is a smell of sulfur, the oceans are boiling, and there are rotten eggs?

$$P(+m \mid +s, +b, +e) =$$

Bayesian network construction



Bayesian network: Scenarios

- **Scenario 1:** Network structure **known** and all variables **observable**
 - Compute only the CPT entries
- **Scenario 2:** Network structure **known** while some variables **hidden**
 - Gradient descent (greedy hill-climbing) method, i.e., search for a solution along the steepest descent of a criterion function
- **Scenario 3:** Network structure **unknown**, all variables **observable**
 - Search through the model space to reconstruct network topology
- **Scenario 4:** Network structure **unknown** and all variables **hidden**
 - No good algorithms known for this purpose

Reference: D. Heckerman. A Tutorial on Learning with Bayesian Networks. In Learning in Graphical Models, M. Jordan, ed.. MIT Press, 1999.

Build a Bayesian network

- Certain conditional independence relationships can guide the knowledge engineer to build the topology of the network.
- The Chain rule holds for any set of random variables.

$$\begin{aligned}\mathbf{P}(x_1, \dots, x_n) &= \prod_{i=1}^n \mathbf{P}(x_i | x_{i-1}, \dots, x_1) \\ &= P(x_n | x_{n-1}, \dots, x_1) P(x_{n-1}, \dots, x_1) \\ &= P(x_n | x_{n-1}, \dots, x_1) P(x_2 | x_{n-2}, \dots, x_1) \dots P(x_2 | x_1) P(x_1)\end{aligned}$$

- We generally assert that, for every variable X_i in the network

$$\mathbf{P}(X_i | X_{i-1}, \dots, X_1) = \mathbf{P}(X_i | X_{Parent(i)}) *$$

where $Parent(i) \subseteq \{X_{i-1}, \dots, X_1\}$.

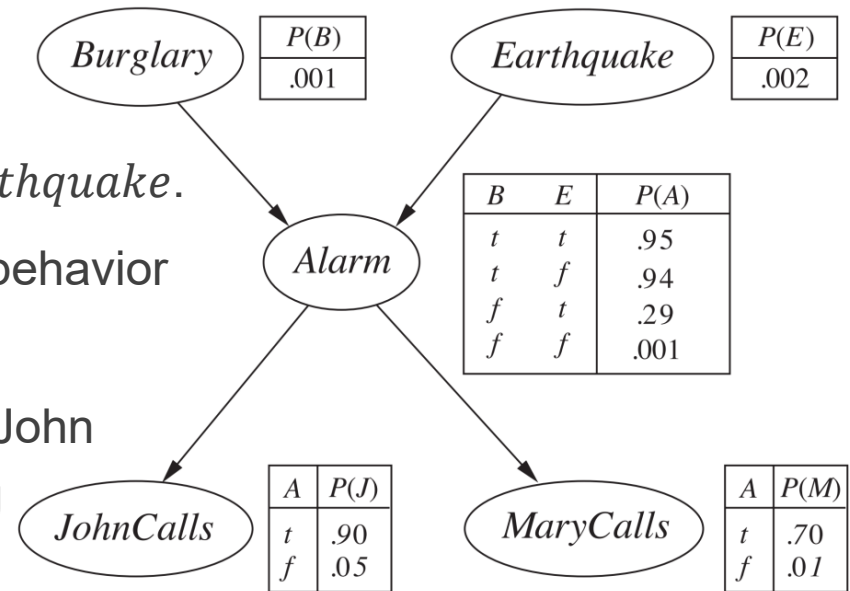
Construct a Bayesian network

- Each node must be **conditionally independent** of its other **predecessors** in the node ordering, **given its parents**.
- **Nodes:** Identify the set of variables required to model the domain and order them, $\{X_1, \dots, X_n\}$.
 - Any order will work, but the resulting network will be more compact if **the variables are ordered such that causes precede effects**.
- **Links:** For $i = 1$ to n do:
 - Choose, from $\{X_1, \dots, X_{i-1}\}$, a **minimal set of parents for X_i** such that Equation ***** is satisfied.
 - For each parent insert a link from the parent to X_i .
 - CPTs: Write down the conditional probability table, $P(X_i \mid X_{Parent(i)})$.

Construct a Bayesian network

- Intuitively, the **parents of node X_i** should contain all those nodes in $\{X_1, \dots, X_{i-1}\}$ that **directly influence X_i** .

- MaryCalls* is indirectly influenced by whether there is a *Burglary* or an *Earthquake*.
- These events influence Mary's calling behavior only through their effect on the *Alarm*
- Given the state of the *Alarm*, whether John calls has no influence on Mary's calling



- That is,

$$P(\text{MaryCalls} \mid \text{JohnCalls}, \text{Alarm}, \text{Earthquake}, \text{Burglary}) = P(\text{MaryCalls} \mid \text{Alarm})$$

- Thus, *Alarm* will be the only parent node for *MaryCalls*.

Construct a Bayesian network

- The network is guaranteed to be **acyclic**.
 - Each node is connected only to earlier nodes.
- Bayesian networks contain **no redundant probability values**.
 - If there is no redundancy, then there is no chance for inconsistency.
- It is impossible for the domain expert to create a Bayesian network that violates the axioms of probability.

Cas studies: Fire diagnosis

- You want to diagnose whether there is a fire in a building
- You can receive reports (possibly noisy) about whether everyone is leaving the building
- If everyone is leaving, this may have been caused by a fire alarm.
- If there is a fire alarm, it may have been caused by a fire or by tampering.
- If there is a fire, there may be smoke.



Fire diagnosis: Define variables

- Start by choosing the random Boolean variables for this domain
- *Tampering (T)*: the alarm has been tampered with
- *Fire (F)*: there is a fire
- *Alarm (A)*: there is an alarm
- *Smoke (S)*: there is smoke
- *Leaving (L)*: there are lots of people leaving the building
- *Report (R)*: the sensor reports that everyone are leaving the building

Fire diagnosis: Chain rule

- Define a total ordering of variables
 - Choose an order that follows the causal sequence of events
 - Fire (F) Tampering (T) Alarm (A) Smoke (S) Leaving (L) Report (R)
- Consider the following chain rule and use given clues to simplify it

$$\mathbf{P}(F, T, A, S, L, R) = \mathbf{P}(F) \mathbf{P}(T \mid F) \mathbf{P}(A \mid F, T) \mathbf{P}(S \mid F, T, A) \\ \mathbf{P}(L \mid F, T, A, S) \mathbf{P}(R \mid F, T, A, S, L)$$

Fire diagnosis: Build a topology

- *Fire* (F) is the first variable in the ordering, X_1 , which has no parent.

$\mathbf{P}(F) \mathbf{P}(T | F) \mathbf{P}(A | F, T) \mathbf{P}(S | F, T, A) \mathbf{P}(L | F, T, A, S) \mathbf{P}(R | F, T, A, S, L)$



- *Tampering* (T) is independent of fire
 - Learning that one is true/false would not change your beliefs about the probability of the other.

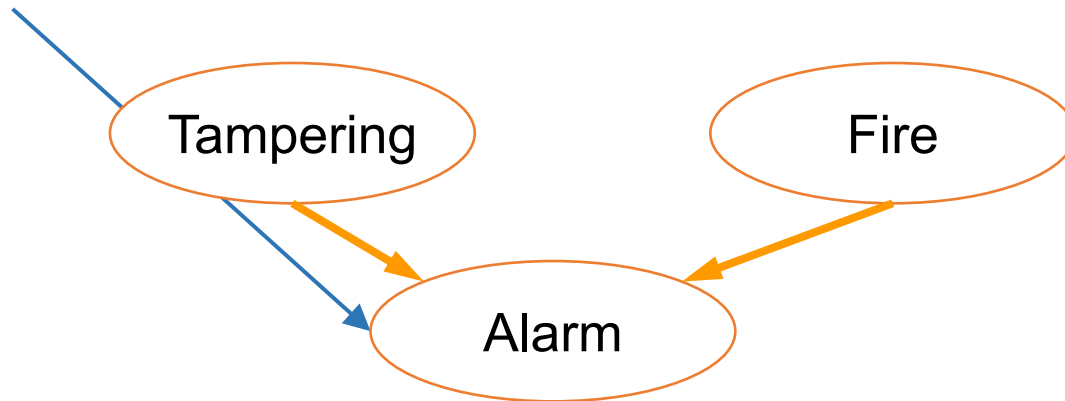
$\mathbf{P}(F) \mathbf{P}(T) \mathbf{P}(A | F, T) \mathbf{P}(S | F, T, A) \mathbf{P}(L | F, T, A, S) \mathbf{P}(R | F, T, A, S, L)$



Fire diagnosis: Build a topology

- *Alarm* (A) depends on both *Fire* and *Tampering*: it could be caused by either or both.

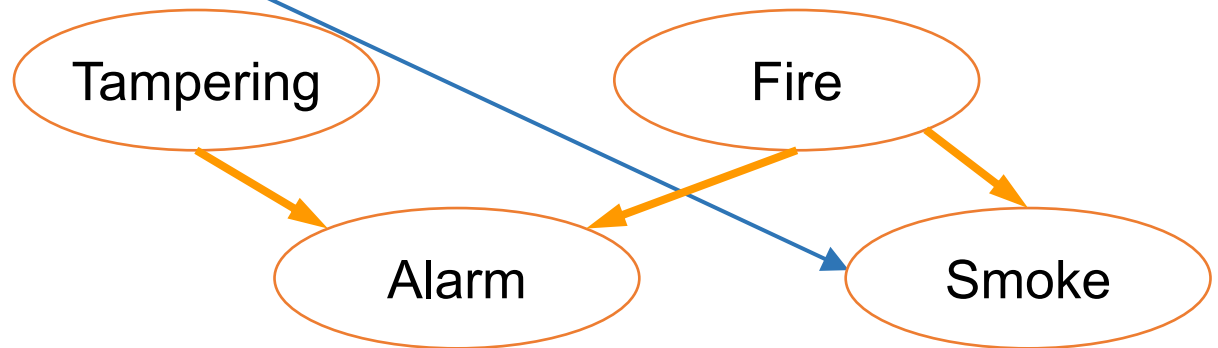
$P(F)$ $P(T)$ $P(A \mid F, T)$ $P(S \mid F, T, A)$ $P(L \mid F, T, A, S)$ $P(R \mid F, T, A, S, L)$



Fire diagnosis: Build a topology

- *Smoke* (S) is caused by *Fire*, and so is independent of *Tampering* and *Alarm*, given whether there is a *Fire*.

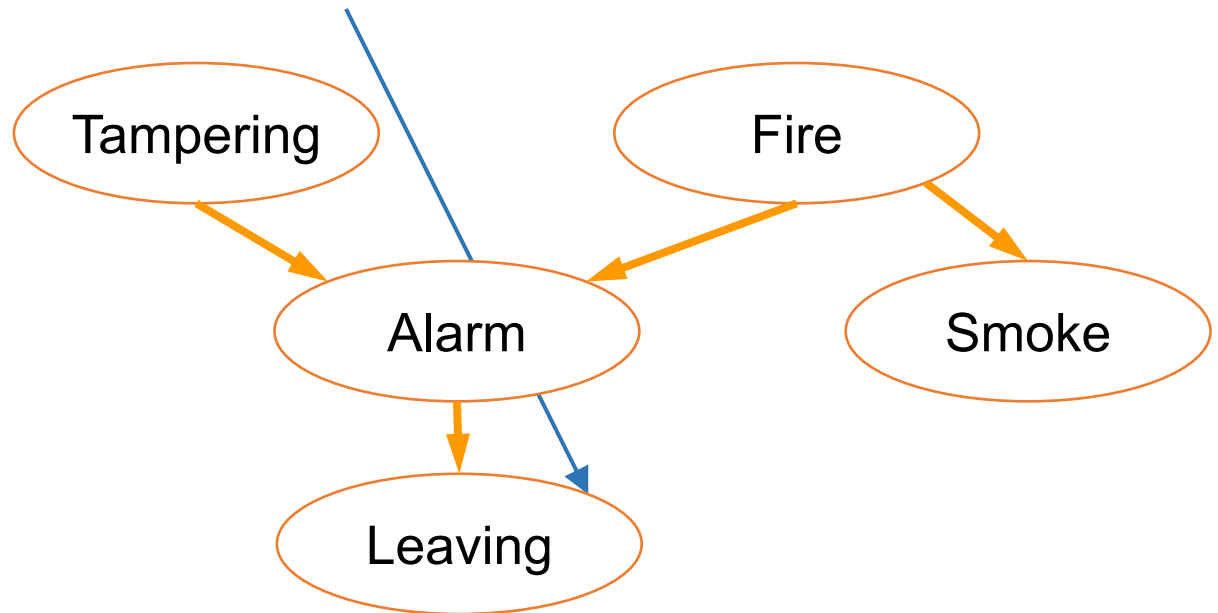
$$\mathbf{P}(F) \mathbf{P}(T) \mathbf{P}(A \mid F, T) \mathbf{P}(S \mid F) \mathbf{P}(L \mid F, T, A, S) \mathbf{P}(R \mid F, T, A, S, L)$$



Fire diagnosis: Build a topology

- *Leaving* (L) is caused by *Alarm*, and thus is independent of the other variables, given *Alarm*.

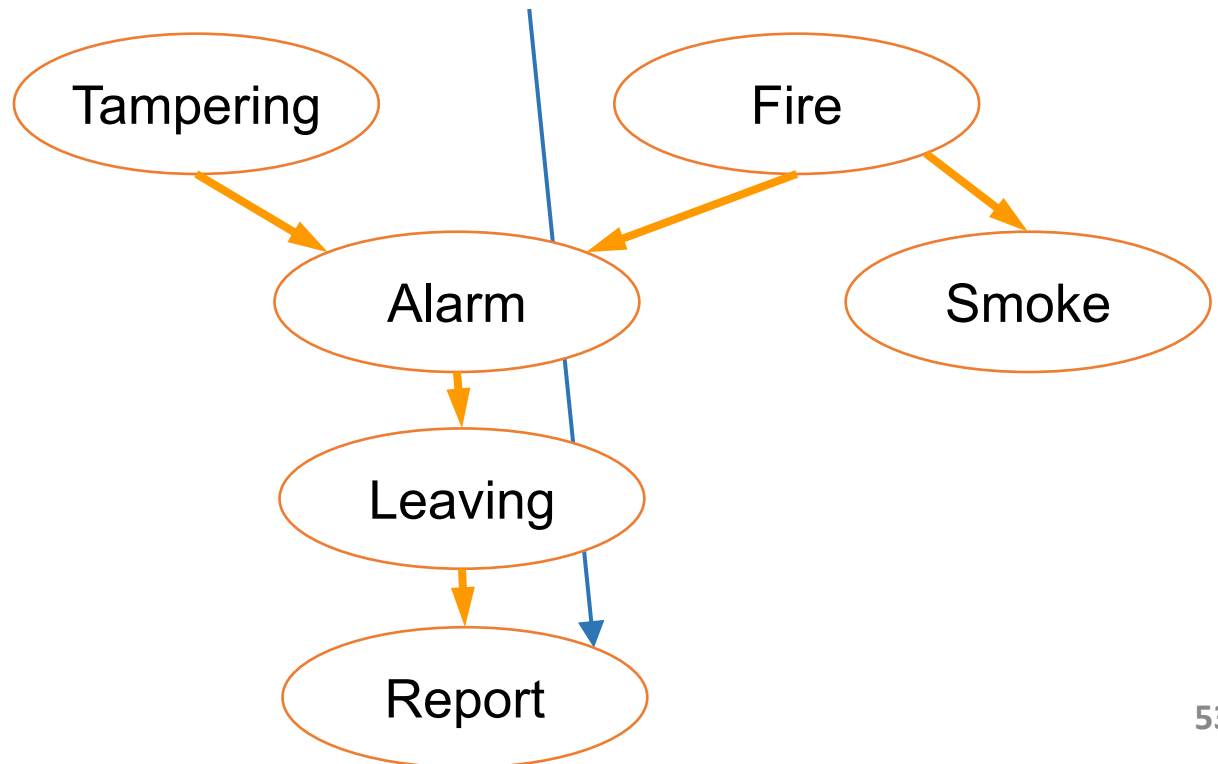
$\mathbf{P}(F)$ $\mathbf{P}(T)$ $\mathbf{P}(A \mid F, T)$ $\mathbf{P}(S \mid F)$ $\mathbf{P}(L \mid \mathbf{A})$ $\mathbf{P}(R \mid F, T, A, S, L)$



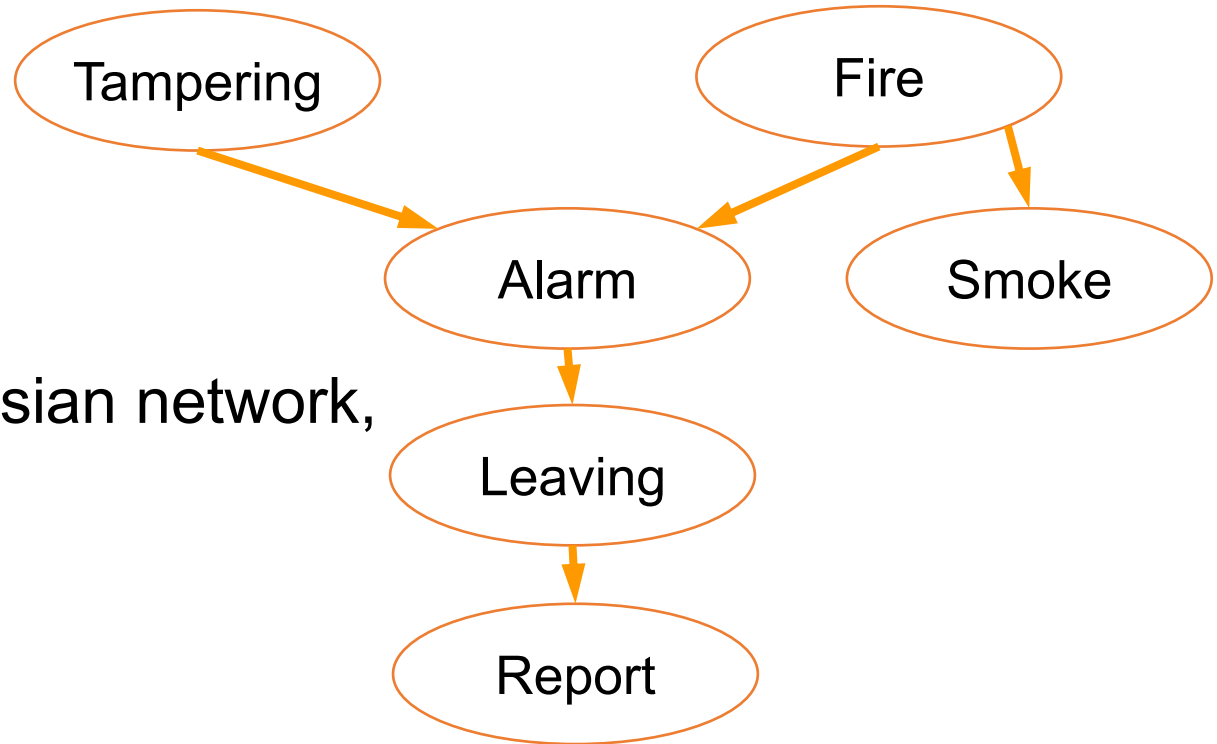
Fire diagnosis: Build a topology

- *Report (R)* is caused by *Leaving*, and thus is independent of the other variables given *Leaving*

$P(F)$ $P(T)$ $P(A | F, T)$ $P(S | F)$ $P(L | A)$ $P(R | L)$



Fire diagnosis: Build a topology



- The resulting Bayesian network, and

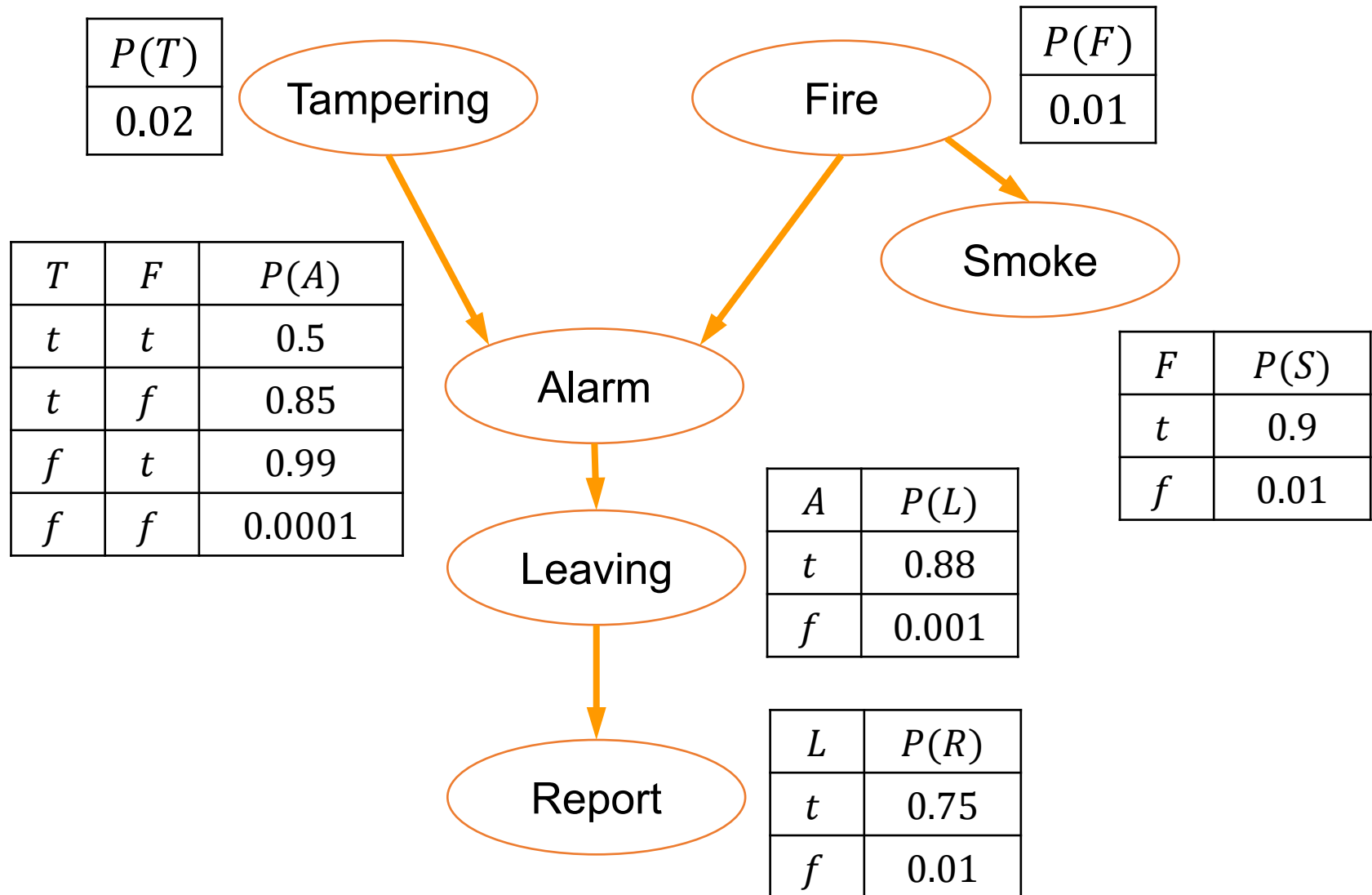
- The corresponding **compact factorization** of the original FJPD

$$\mathbf{P}(F, T, A, S, L, R) = \mathbf{P}(F) \mathbf{P}(T) \mathbf{P}(A | F, T) \mathbf{P}(S | F) \mathbf{P}(L | A) \mathbf{P}(R | L)$$

Fire diagnosis: Specify CPTs

- How many probabilities do we need to specify for this Bayesian network?
- How many probabilities do we explicitly specify for *Fire*?
- How many probabilities do we explicitly specify for *Alarm*?

Fire diagnosis: Specify CPTs



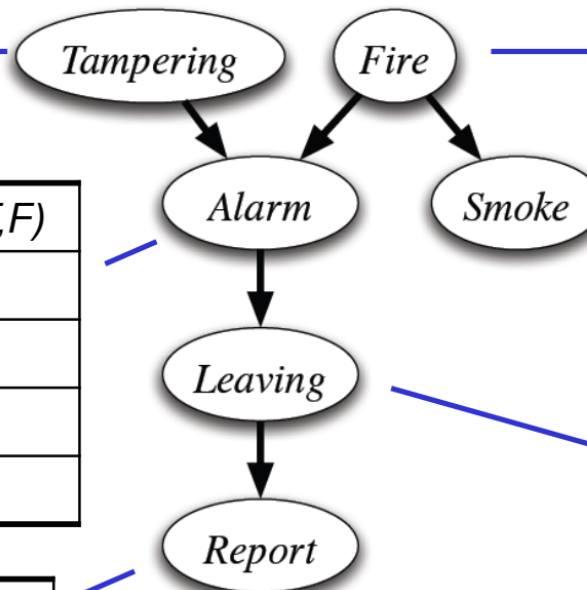
Fire diagnosis: Calculations

- $P(T = t, F = f, A = t, S = f, L = t, R = t) = ?$
- $P(T = t) \times P(F = f) \times P(A = t | T = t, F = f) \times P(S = f | F = f)$
 $\times P(L = t | A = t) \times P(R = t | L = t)$

$P(\text{Tampering}=t)$
0.02

Tampering T	Fire F	$P(\text{Alarm}=t T,F)$
t	t	0.5
t	f	0.85
f	t	0.99
f	f	0.0001

Leaving	$P(\text{Report}=t L)$
t	0.75
f	0.01



$P(\text{Fire}=t)$
0.01

Fire F	$P(\text{Smoke}=t F)$
t	0.9
f	0.01

Alarm	$P(\text{Leaving}=t A)$
t	0.88
f	0.001

$$= 0.02 \times (1 - 0.01) \times 0.85 \times (1 - 0.01) \times 0.88 \times 0.75$$

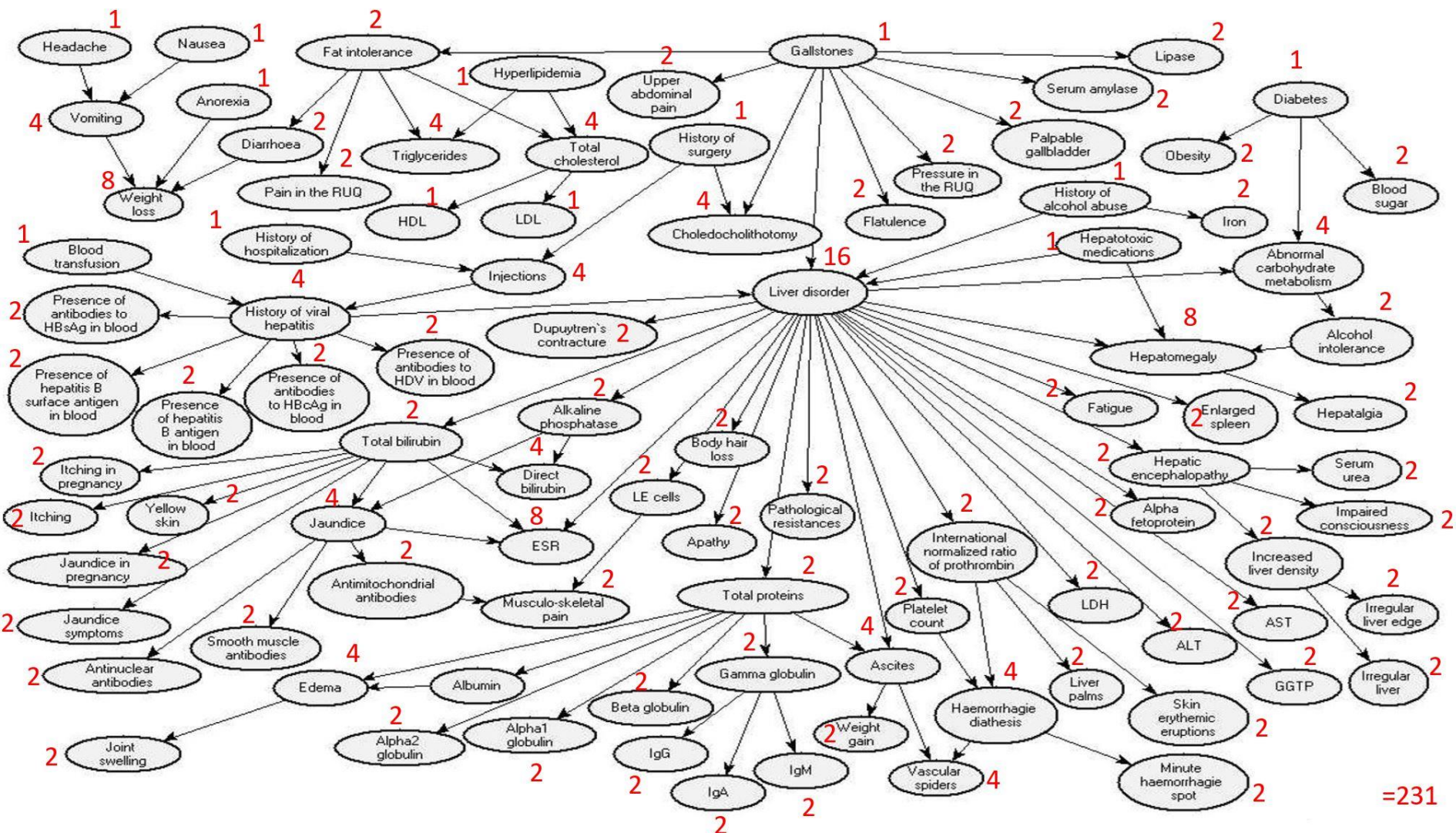
$$= \mathbf{0.126}$$

Fire diagnosis: Specify CPTs

- How many probabilities do we need to specify for this Bayesian network?
- $P(Tampering)$: 1 probability $P(T = t)$
- $P(Alarm | Tampering, Fire)$: 4 (independent)
 - 1 probability for each of the 4 instantiations of the parents
- For all other variables with only one parent: 2 probabilities: one for the parent being true and one for otherwise
- In total: $1+1+4+2+2+2 = 12$ (compared to $2^6-1=63$ for FJPD)

Bayesian networks vs. FJPD

- A CPT for a Boolean variable X_i with k Boolean parents has 2^k rows for the combinations of parent values.
- If each variable has no more than k parents, the complete network requires to specify $n2^k$ numbers.
 - For $k \ll n$, this is a substantial improvement.
 - The numbers required grow linearly with n , vs. $O(2^n)$ for the FJPD.
- For example, a Bayesian network with 30 Boolean variables, each with 5 parents, needs 30×2^5 probabilities.
 - Meanwhile, a JPD requires 2^{30} probabilities.



~60 nodes, max 4 parents per node

Need $\sim 60 \times 2^4 = 15 \times 2^6$ probabilities instead of 2^{60} probabilities for the FJPD

Bayesian networks vs. JPD

- What happens if the network is fully connected, or $k \approx n$?
 - Not much saving compared to the numbers needed for FJPD.
- Bayesian networks are useful in sparse domains (or locally structured domains).
 - Sparse domain = a domain in which each component interacts with (is related to) a small fraction of other components
- What if this is not the case in a domain we reason about?
 - We may need to make simplifying assumptions to reduce the dependencies in a domain.

Where do the CPTs come from?

- From experts: tedious, costly, not always reliable
- From data: **Machine Learning**
 - There are algorithms to learn both structures and numbers.
 - It can be hard to get enough data.
- Still, usually better than specifying the FJPD.

Acknowledgements

- This slide is mainly based on the textbook AIMA (3rd edition)
- Some parts of the slide are adapted from
 - Cristina Conati, *Lecture 20: Bayesian Networks: Construction*, Computer Science CPSC 322, University of British Columbia.
 - Charikar and Sadigh, *Lecture 13: Bayesian networks I*, CS221, Spring 2019, Stanford University.



...the end.

